

Experiment No. 3

To perform no load & blocked rotor test on 3-phase Induction motor.

Experiment No- 3

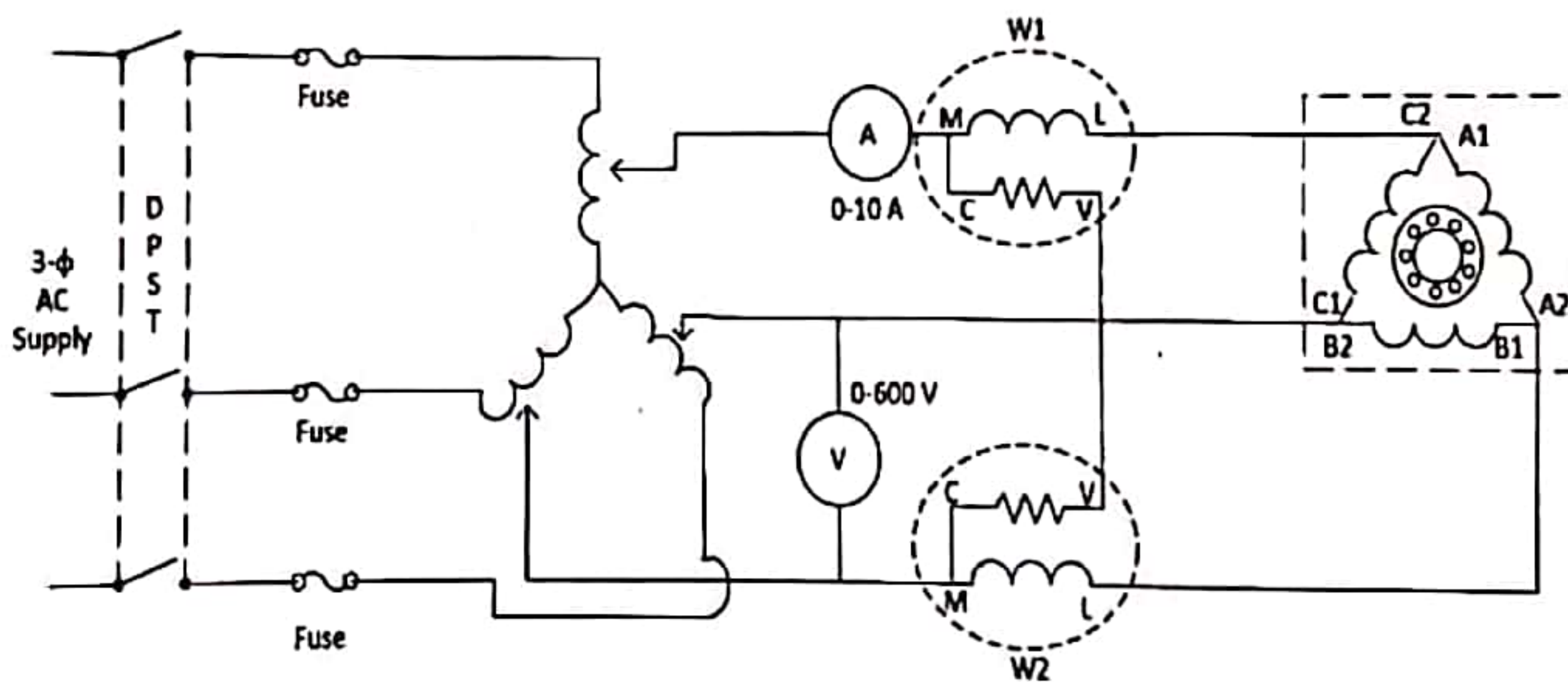
Aim:

To perform no load and blocked rotor test on 3 phase Induction Motor .

Apparatus Required:

S. No.	Apparatus/Accessories	Type	Range	Quantity
1	Ammeter	MI	0 - 10/20A	1
2	Voltmeter	MI	0 - 300/600V	1
3	Wattmeter	-	10/20A 200/400V	2
4	3 phase variac	-	15A 400/0-400V	1

Circuit Diagram:



Theory:

The performance and other characteristics curve of a 3-phase induction motor may be obtained by testing the machine under load. However, the test under actual loading is in many cases difficult to carry out and in some cases even impracticable as for instance with large capacity machines built in electrical manufacturing industries in such cases characteristics can be obtained with the help of a circle diagram based on relatively simple on no load and blocked rotor test.

The circle diagram is based on the approximate equivalent circuit of the induction motor (where the no load admittance branch is shifted to the supply terminals) and represents the locus of the primary current vector of the machine.

No load test:

The no load test is similar to the open circuit test on a transformer. It is performed to obtain the magnetizing branch parameters (shunt parameters) in the induction machine equivalent circuit. In this test, the motor is allowed to run with no-load at the rated voltage of rated frequency across its terminals. Machine will rotate at almost synchronous speed, which makes slip nearly equal to zero. This causes the equivalent rotor impedance to be very large (theoretically infinite neglecting the frictional and rotational losses). Therefore, the rotor equivalent impedance can be considered to be an open circuit.

If i_0 , V_0 , W_{01} , W_{02} are the readings during no load test then,

$$\tan\phi_0 = \sqrt{3}(W_{01} - W_{02}) / (W_{01} + W_{02})$$

Thus, no load power factor angle can be calculated.

Blocked rotor test:

Blocked rotor test is similar to the short circuit test on a transformer. It is performed to calculate the series parameters of the induction machine i.e., its leakage impedances. The rotor is blocked to prevent rotation and balanced voltages are applied to the stator terminals at a frequency of 25 percent of the rated frequency at a voltage where the rated current is achieved. Under the reduced voltage condition and rated current, core loss and magnetizing component of the current are quite small percent of the total current.

If i_{sc} , V_{sc} , W_{sc1} , W_{sc2} are the readings during blocked rotor test, then

$$\tan\phi_{sc} = \sqrt{3}(W_{sc1} - W_{sc2}) / (W_{sc1} + W_{sc2})$$

Thus, short circuit power factor can be calculated.

The power factor of the motor is quite low under both test conditions. Thus one wattmeter in the circuit will give negative reading, which may be recorded by reversing the terminals of pressure or current coil.

Procedure:

Steps to perform the experiment.

1. Connections are made as shown in the diagram.
2. The motor is started on no load (with belt loose) with the aid of a three phase to variac.
3. The applied voltage to the unloaded three phase induction motor is adjusted to rated voltage and readings of Input power, current and voltage are recorded .Switch off the supply to the motor.
4. The connections for the block rotor test are essentially the same as those of the no load test except the ammeter and wattmeter are chosen to be able to as the rated current of the motor.
5. In block rotor test the rotor is held firmly so that it does not rotate. A reduced voltage is applied with the help of a three phase variac so that rated current flow in the stator winding and readings of input power current and voltage are recorded.
6. The dc resistance of stator winding per phase is measured by voltmeter ammeter method and its per phase effect value is calculated from the relation.
7. Effective resistance= 1.8 times dc resistance of one phase.

Observations:**No load test:**

S. No.	voltage(v)	current(i)	power		$\cos\phi$
			W1	W2	
1.					

Blocked rotor test :

S. No.	voltage(v)	current(i)	power		$\cos\phi$
			W1	W2	
1.					

Calculation: (Do Calculations after precautions, if required)

Result:

The Power factor calculated after conducting the Block Rotor test is

The Power factor calculated after conducting the No Load test is

Precautions:

1. Slowly increase the voltage and note the readings carefully.
2. Connect the ammeter and voltmeter with correct polarity.
3. Multiplying factor of wattmeter should be checked properly
4. Loading should be done slowly and precisely by rotating the spring valve.
5. All connections should be tight.
6. The rotor should be blocked properly at the starting of the motor.
7. It should be kept in mind that the applied voltage on the stator terminals should be low otherwise normal voltage could damage the winding of the stator.